

Explicit solutions of the shallow water equations

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The depth relative to the bed is h , the bed elevation is z , and the absolute elevation is η , so $h = \eta - z$. The depth-averaged velocity is u . Gravity is g , and $c = \sqrt{gh}$.

1. The equations

On a flat frictionless bed the one-dimensional system is

$$\begin{aligned} h_t + (hu)_x &= 0, \\ u_t + uu_x + gh_x &= 0. \end{aligned} \tag{1}$$

With a bed $z(x)$ the second equation of the above changes to

$$u_t + uu_x + g\eta_x = 0. \tag{2}$$

With linear friction on the bed of coefficient $\tau > 0$ it is further changed to

$$u_t + uu_x + g\eta_x = -\tau u. \tag{3}$$

Writing the equations in matrix form we have

$$\begin{pmatrix} h \\ u \end{pmatrix}_t + \begin{pmatrix} u & h \\ g & u \end{pmatrix} \begin{pmatrix} h \\ u \end{pmatrix}_x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{4}$$

The characteristic polynomial set to zero of the defining matrix is $(\lambda - u)^2 = gh$, thus the characteristic speeds are $u \pm c$ and the Riemann invariants are $R_{\pm} = u \pm 2c$.

2. Explicit solutions

2.1. Stable resting state

For arbitrary bed $z(x)$ (possibly $z(x) \equiv 0$), $u = 0$ and $\eta = H$ is a trivial solution. Since z, u, η , and therefore h are all constant all the derivatives are 0, thus the equations are trivially satisfied.

2.2. Steady flow

Bernoulli's law tells us the following

$$\frac{u^2}{2} + g(h + z) = B = \text{const}. \tag{5}$$

And let the discharge on one side be $q_0 = hu$, with q_0 constant.

Given a positive depth $h(x)$ the rest is explicit:

$$u(x) = \frac{q_0}{h(x)}, \quad z(x) = \frac{B}{g} - h(x) - \frac{q_0^2}{2gh(x)^2}. \tag{6}$$

After some calculations one can check that this is indeed a solution to the equations for a sufficiently smooth $h(x)$. For a $h(x)$ that is not smooth enough there may be a jump between two states, but then the Rankine–Hugoniot conditions can pin down the solution.

2.3. Dry dam break (Ritter)

The starting conditions are:

$$h(x, 0) = \begin{cases} h_L & x < 0 \\ 0 & x > 0 \end{cases} \quad u(x, 0) = 0. \quad (7)$$

Let $c_L = \sqrt{gh_L}$, then the solution is

$$h(x, t) = \begin{cases} h_L & \text{if } x < -c_L t \\ \frac{1}{9g}(2c_L - \frac{x}{t})^2 & \text{if } -c_L t \leq x \leq 2c_L t \\ 0 & \text{if } x > 2c_L t \end{cases} \quad (8)$$

$$u(x, t) = \begin{cases} 0 & \text{if } x < -c_L t \\ \frac{2}{3}(c_L + \frac{x}{t}) & \text{if } -c_L t \leq x \leq 2c_L t \\ 0 & \text{if } x > 2c_L t. \end{cases} \quad (9)$$

2.4. Wet dam break (Stoker)

The starting conditions are:

$$h(x, 0) = \begin{cases} h_L & x < 0 \\ h_R & x > 0 \end{cases} \quad u(x, 0) = 0, \quad (10)$$

with $0 < h_R < h_L$. That is, the only thing that changed compared to the dry dam break case is that on the right side of the dam there is water.

Let $c_L = \sqrt{gh_L}$ and $c_R = \sqrt{gh_R}$ and let $u_m = 2(c_L - c_m)$ and $c_m = \sqrt{gh_m}$, where h_m is the solution to

$$u_m = 2(c_L - c_m) = (h_m - h_R)\sqrt{\frac{g(h_m + h_R)}{2h_m h_R}}. \quad (11)$$

It can be shown that such an h_m exists.

Introduce the notations $\xi = x/t$ and $s = \frac{h_m u_m}{h_m - h_R}$. With these notations the solution is of the form

$$(h, u) = \begin{cases} (h_L, 0) & \text{if } \xi < -c_L \\ \left(\frac{1}{9g}(2c_L - \xi)^2, \frac{2}{3}(c_L + \xi)\right) & \text{if } -c_L \leq \xi \leq u_m - c_m \\ (h_m, u_m) & \text{if } u_m - c_m \leq \xi < s \\ (h_R, 0) & \text{if } \xi > s. \end{cases} \quad (12)$$

2.5. Simple wave

A simple wave is a wave for which all but one Riemann invariants are constnat.

First note that the original equation can be rewritten in terms of $c = \sqrt{gh}$:

$$\begin{cases} 2c_t + 2uc_x + cu_x = 0 \\ u_t + uu_x + 2cc_x = 0 \end{cases} \quad (13)$$

Let $u + 2c = m$ where m is constant, that is, $u = m - 2c$. For any given differentiable φ (where $\varphi'(c) \neq 3t$) let $c(x, t)$ be the solution to

$$x = (m - 3c)t + \varphi(c). \quad (14)$$

From this relation we can calculate the partial derivatives of c :

$$c_x = \frac{1}{\varphi'(c) - 3t}, \quad c_t = -(m - 3c)c_x = -(u - c)c_x. \quad (15)$$

Rearranging the equation for c_t we get that $c_t + (u - c)c_x = 0$. We can express u_x and u_t in terms of c_t and c_x aswell:

$$u_t = -2c_t, \quad u_x = -2c_x. \quad (16)$$

Checking the mass equation in the shallow water equation written in terms of c we get that

$$2c_t + 2uc_x + cu_x = 2c_t 2uc_x - 2cc_x = 2(c_t + (u - c)c_x) = 0, \quad (17)$$

because the expression inside the paranthesis was shown to be 0.

For the momentum equation we can carry out a similar calculation:

$$u_t + uu_x + 2cc_x = -2c_t - 2uc_x + 2cc_x = -2(c_t + (u - c)c_x) = 0. \quad (18)$$

In summary, if we fix any differentiable φ and a constant m , then solve the equation for c we can calculate h as $h = c^2/g$ and $u = m - 2c$.

The same argument can be said if we fix the other Riemann invariant to be constant, with $u - 2c = k$, then we need to use the other characteristic speed $u + c$ and solve for c the equation $x = (u + c)t + \varphi(c)$. Here the condition is that $\varphi'(c) \neq -3t$.

Note, that for example Ritter's dry dam break is a special case of this.

3. Example plots

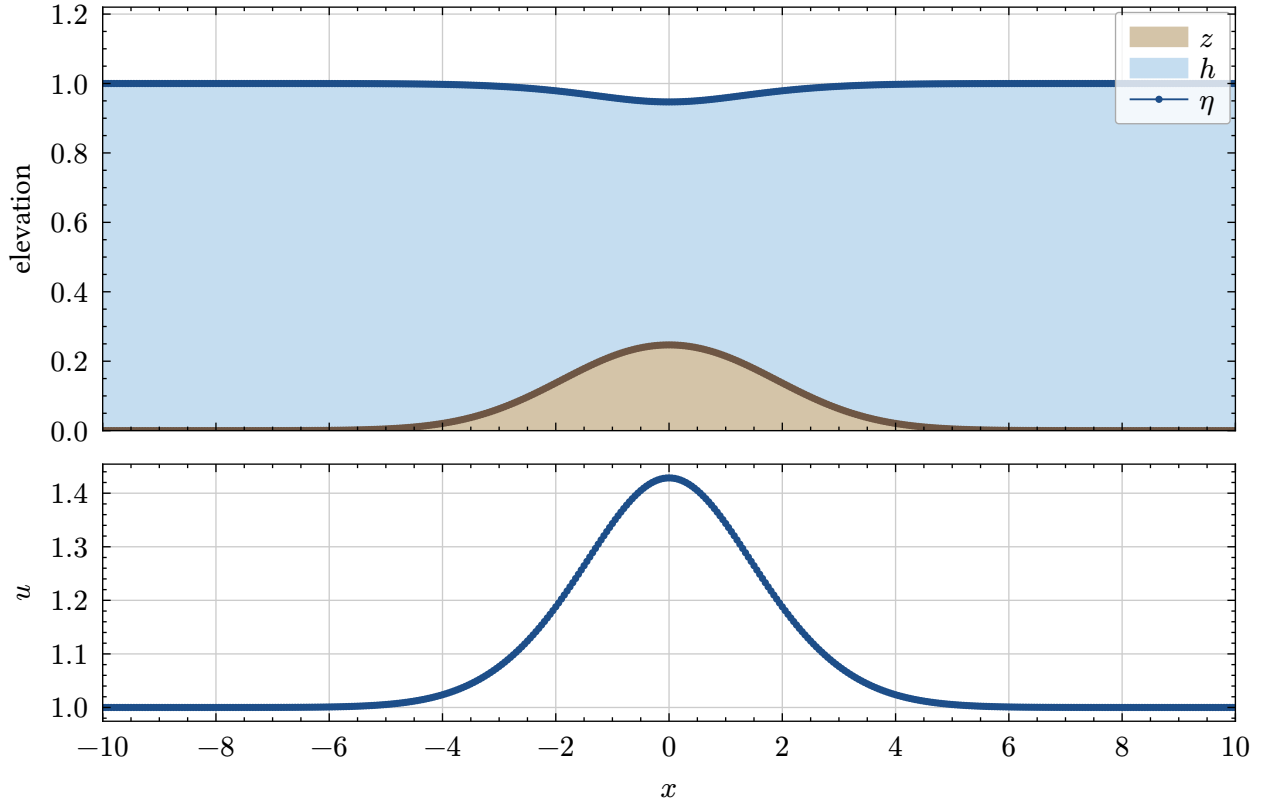


Figure 1: Bed, water, and free surface for this depth, and the velocity $u = q_0/h$.

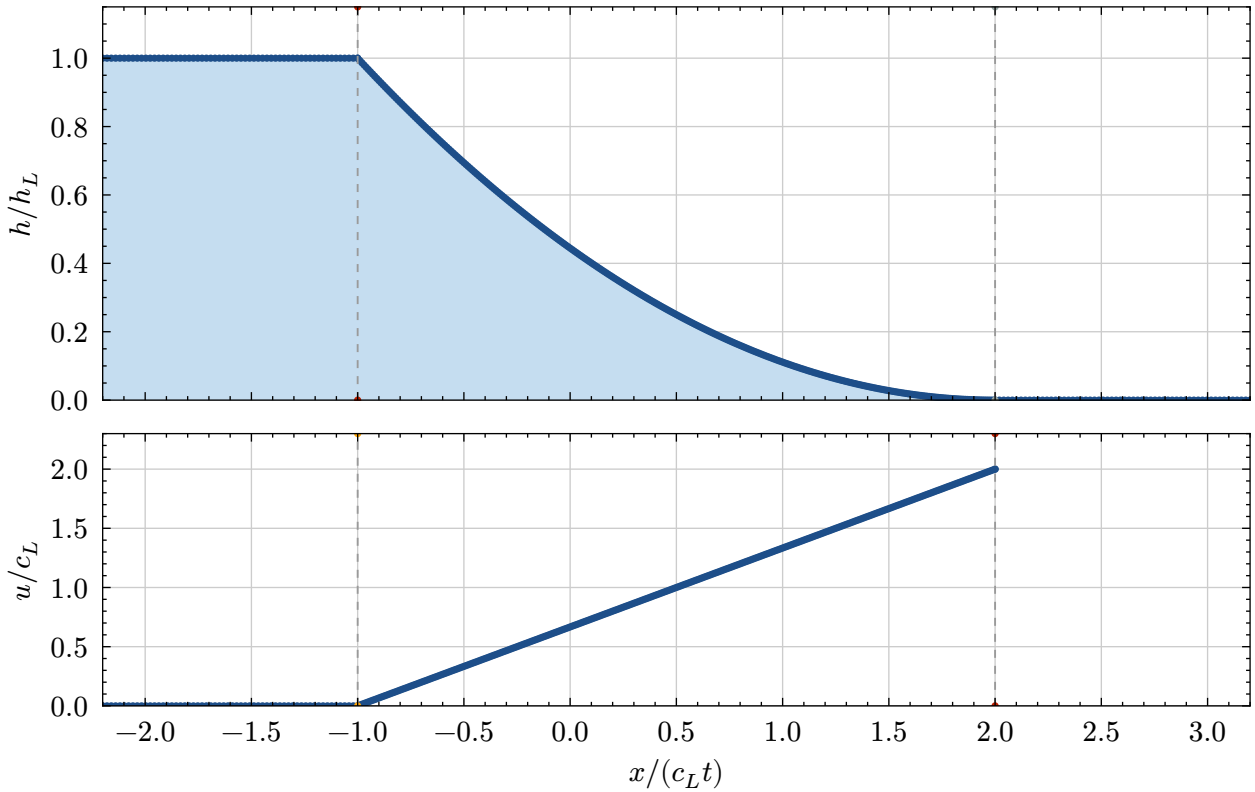


Figure 2: Ritter's solution at a fixed $t > 0$. The dashed lines are $x = -c_L t$ and the dry front $x = 2c_L t$.

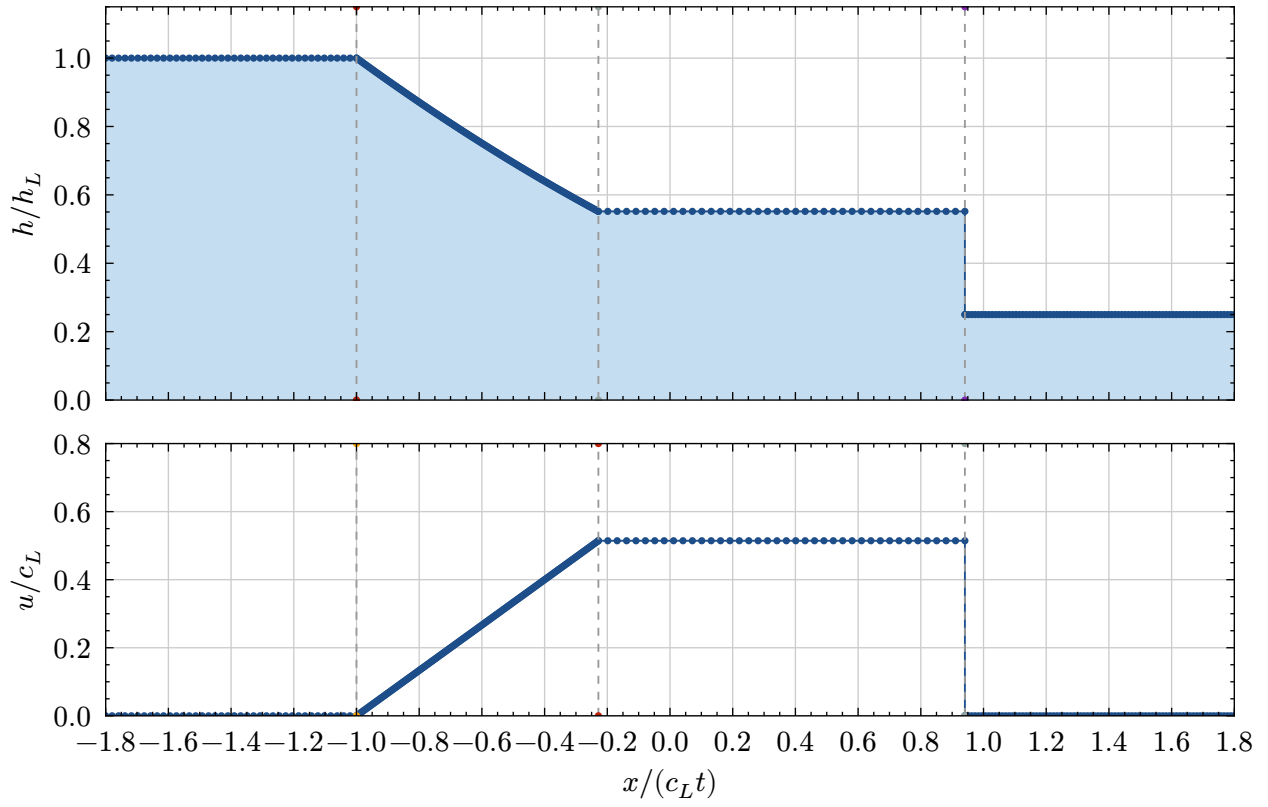


Figure 3: Stoker's solution for $h_R = h_L/4$ at a fixed $t > 0$. The dashed lines are $x = -c_L t$, the downstream edge of the rarefaction, and the shock $x = st$.

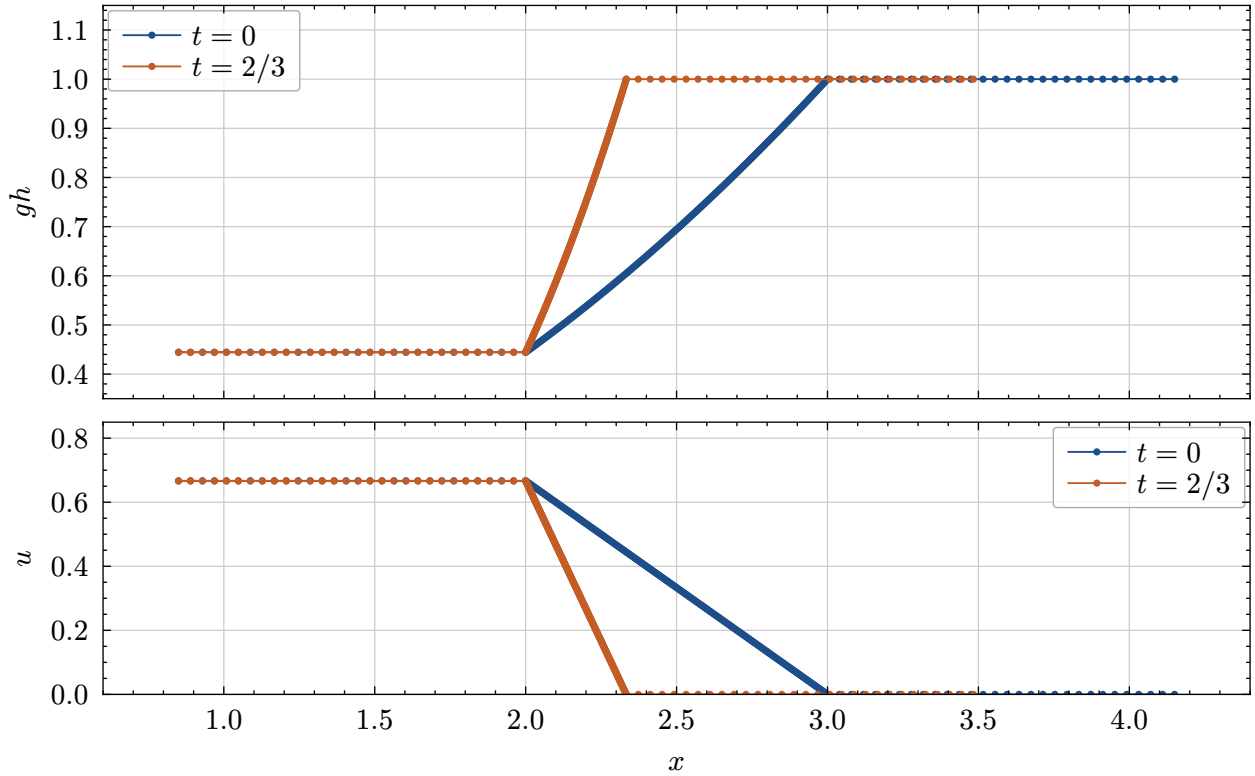


Figure 4: A compressive simple wave of the first family, before breaking at $t = 1$. Depth is drawn as $gh = c^2$.